JEE Advanced Exam 2021
(Paper & Solution)

PAPER-1

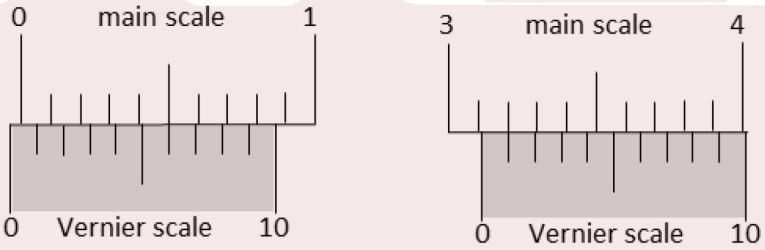
PHYSICS

SECTION – 1

- This section contains **FOUR (04)** questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	: **+3**	If ONLY the correct option is chosen.
Zero Marks	: **0**	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: **–1**	In all other cases.

Q.1 The smallest division on the main scale of a Vernier calipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this calipers with no gap between its two jaws. The figure on the right shows the reading with solid sphere held between the jaws. The correct diameter of the sphere is-



(A) 3.07 cm (B) 3.11 cm (C) 3.15 cm (D) 3.17 cm

Ans. **[C]**

Sol. 9 MSD = 10 VSD

L.C. = 0.1 – 0.9 MSD = 0.01 cm

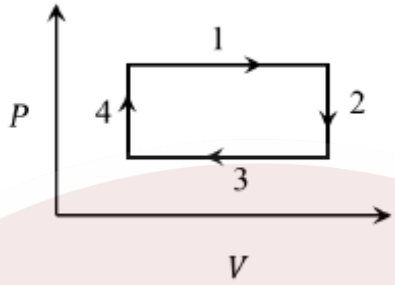
when there is no object, negative zero error.

Zero error = –0.04 cm

Observed reading (without error) = $3.1 + 1 \times 0.01 = 3.11$ cm

Actual reading (with error) = $3.11 - (-0.04) = 3.15$ cm

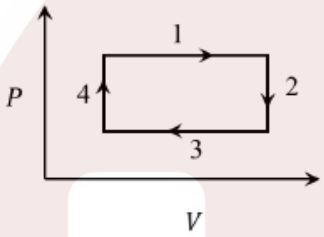
Q.2 An ideal gas undergoes a four step cycle as shown in the P-V diagram below. During this cycle, heat is absorbed by the gas in



(A) steps 1 and 2 (B) steps 1 and 3 (C) steps 1 and 4 (D) steps 2 and 4

Ans. **[C]**

Sol.



$1 \rightarrow$ isobaric process as V increase $\Rightarrow$ Temperature also increase

ΔT = +ve gas absorbs heat.

$2 \rightarrow$ isobaric process

$P \propto T \Rightarrow$ P decrease = T decreases

heat rejected by gas.

$3 \rightarrow$ isobaric process

$V \propto T \Rightarrow$ V decrease = T decreases

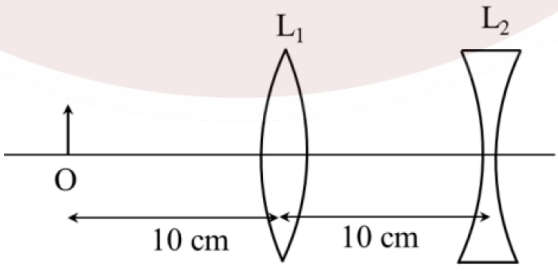
heat rejected by gas.

$4 \rightarrow$ isobaric process

$P \propto T \Rightarrow$ P increase = T increase

heat absorbed by gas.

Q.3 An extended object is placed at point O, 10 cm in front of a convex lens L_1 and a concave lens L_2 is placed 10 cm behind it, as shown in the figure. The radii of curvature of all the curved surfaces in both the lenses are 20 cm. The refractive index of both the lenses is 1.5. The total magnification of this lens system is-



(A) 0.4 (B) 0.8 (C) 1.3 (D) 1.6

Ans. **[B]**

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Sol. $\frac{1}{v_1}-\frac{1}{-10}=\frac{1}{20}$

$\frac{1}{v_1}=\frac{1}{20}-\frac{1}{10}$

$v_1 = -20$ cm

$m_1 = \frac{-20}{-10} = 2$

Now I_1 acts as an object for Lens L_2

$u = -30$

$f = -20$

$\frac{1}{v_2}=\frac{-1}{20}-\frac{1}{30} \Rightarrow v_2 = -12$ cm

$m_2 = \frac{-12}{-30} = \frac{2}{5}$

$m_{net} = m_1 \times m_2 = 2 \times \frac{2}{5} = 0.8$

Q.4 A heavy nucleus Q of half-life 20 minutes undergoes alpha-decay with probability of 60% and beta-decay with probability of 40%. Initially, the number of Q nuclei is 1000. The number of alpha-decay of Q in the first one hour is-

(A) 50 (B) 75 (C) 350 (D) 525

Ans. **[D]**

Sol. Remaining nuclei after 1 hrs = $\frac{N_0}{2^3} = \frac{N_0}{8}$

Total decay = $N_0 - \frac{N_0}{8} = \frac{7N_0}{8}$

$\therefore$ Number of α-decay in 1 hour = $\frac{7N_0}{8}\times\frac{60}{100}$

$= \frac{7\times1000\times60}{8\times100} = 525$

SECTION – 2

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme.

  Full Marks : +2 If ONLY the correct numerical value is entered.

  Zero Marks : 0 In all other cases.

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Question Stem for Question Nos.5 and 6

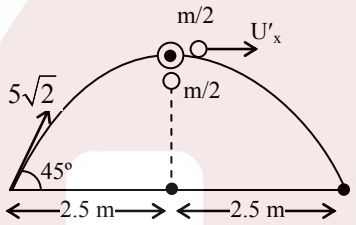
Question Stem

A projectile is thrown from a point O on the ground at angle 45º from the vertical and with a speed $5\sqrt{2}$ m/s. The projectile at the highest point of its trajectory splits into two equal parts. One part falls vertically down to the ground, 0.5 s after the splitting. The other part, t seconds after the splitting, falls to the ground at a distance x meters from the point O. The acceleration due to gravity g = 10 m/s^2.

Q.5 The value of t is _____.
Ans. **[0.5 sec]**

Q.6 The value of x is _____.
Ans. **[7.5 m]**

Sol. for Q.5 & Q.6



$$\text{Range} = \frac{U^2 \sin 2\theta}{g} = \frac{50}{10}$$

R = 5 m

Since there is no external force along x-axis momentum conserved at highest point along x-axis.

$(P_i)_x = (P_f)_x$

$$m\,5 = \frac{m}{2}(0) + \frac{m}{2}U'_x \Rightarrow U'_x = 10 \text{ m/s}$$

$S_n = U'_x t \Rightarrow S_n = 10 \times t$

$$S_y = U'_y t + \frac{1}{2}a_y t^2$$

$$S_y = \frac{U^2 \sin^2\theta}{2g} \Rightarrow \frac{50\times 1}{2\times 0.10\times 2} = \frac{1}{2} \times 10 \times t^2$$

$$t = \frac{1}{2} \Rightarrow t = 0.5 \text{ sec}$$

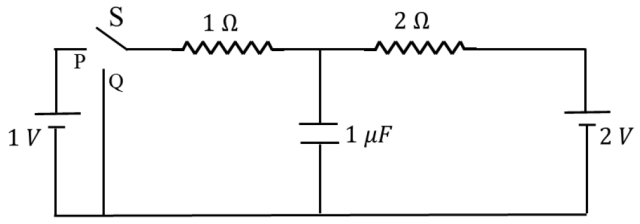
$S_n = 0.5 \times 10 \Rightarrow 5\text{m}$ x = 5 + 2.5

x = 7.5 m

Question Stem for Question Nos.7 and 8

Question Stem

In the circuit shown below, the switch S is connected to position P for a long time so that the charge on the capacitor becomes q_1 μC. Then S is switched to position Q. After a long time, the charge on the capacitor is q_2 μC.

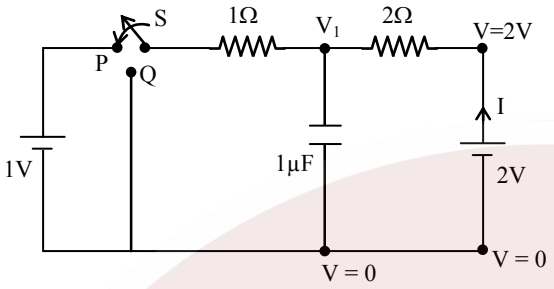


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Q.7 The magnitude of q_1 is _____.

Ans. **[1.33 μC]**

Sol.



When switch 'S' is connected to P for long time

$$I = \frac{1}{3}\ A$$

$V_1 = 2 - 2 \times \frac{1}{3} = \frac{4}{3}$ = P.D. across capacitor

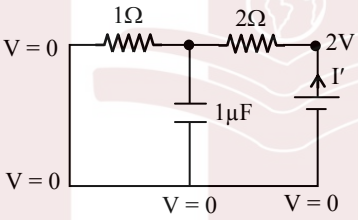
$Q_1 = CV \Rightarrow Q_1 = 1 \times \frac{4}{3} = 1.33\ \mu C$

$q_1 = 1.33\ \mu C$

Q.8 The magnitude of q_2 is _____.

Ans. **[0.67 μC]**

Sol.



$I' = \frac{2}{3} A$

P.D. across capacitor = $2 - 2 \times \frac{2}{3} = \frac{2}{3}$

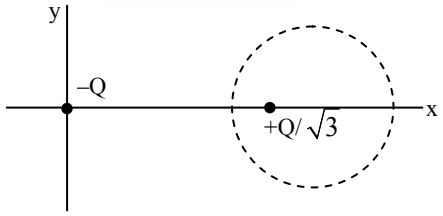
$q_2 = CV$

$\Rightarrow q_2 = 1 \times \frac{2}{3} = 0.67\ \mu C$

Question Stem for Question Nos.9 and 10

Question Stem

Two point charges –Q and $+Q/\sqrt{3}$ are placed in the xy-plane at the origin (0, 0) and a point (2, 0), respectively, as shown in the figure. This results in an equipotential circle of radius R and potential V = 0 in the xy-plane with its center at (b, 0). All lengths are measured in meters.

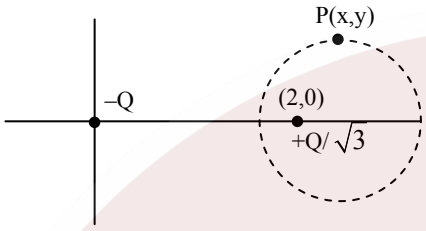


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Q.9 The value of R is _____ meter.
Ans. **[1.73]**

Q.10 The value of b is _____ meter.
Ans. **[3]**

Sol. for Q.9 & Q.10



Potential at point 'P' = 0

V_P due to charge at origin = V_P due to change at (2, 0)

$$\frac{KQ}{\sqrt{x^2+y^2}} = \frac{KQ/\sqrt{3}}{\sqrt{(x-2)^2+y^2}}$$

$$\frac{1}{(x^2+y^2)} = \frac{1}{3(x-2)^2+y^2}$$

$3(x^2 + 4 – 4x + y^2) = x^2 + y^2$

$x^2 – 6x + 6 + y^2 = 0$

$(x – 3)^2 + y^2 = (\sqrt{3})^2$

Centre (3, 0) radius = $\sqrt{3}$

SECTION – 3

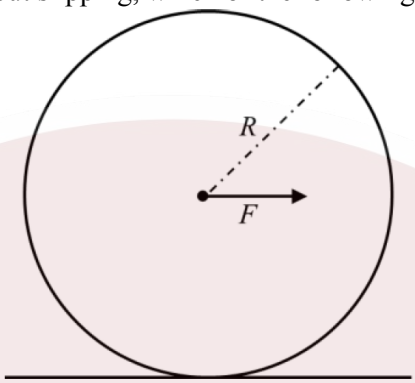
- This section contains **SIX (06)** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme</u> :

Full Marks	: **+4**	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: **+3**	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: **+2**	If three or more options are correct but ONLY two options are chosen, both of which are correct.
Partial Marks	: **+1**	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: **0**	If none of the option is chosen (i.e. the question is unanswered).
Negative Marks	: **–2**	In all other cases.

- For example, in a question (A), (B) and C) and (D) are the ONLY three options corresponding to correct answer, then
  choosing ONLY (A), (B) an (D) will get +4 marks ;
  choosing ONLY (A) and (B) will get +2 marks;
  choosing ONLY (A) and (D) will get +2 marks ;
  choosing ONLY (B) and (D) will get +2 marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option (s) (i.e. the question is unanswered) will get 0 marks and
  choosing any other option (s) will get –2 marks.

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Q.11 A horizontal force F is applied at the center of mass of a cylindrical object of mass m and radius R, perpendicular to its axis as shown in the figure. The coefficient of friction between the object and the ground is μ. The center of mass of the object has an acceleration *a*. The acceleration due to gravity is *g*. Given that the object rolls without slipping, which of the following statement (s) is(are) correct ?



(A) For the same F, the value of *a* does not depend on whether the cylinder is solid or hollow

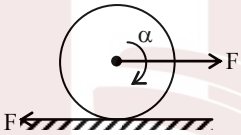
(B) For a solid cylinder, the maximum possible value of *a* is 2μ*g*

(C) The magnitude of the frictional force on the object due to the ground is always *μmg*

(D) For a thin-walled hollow cylinder, $a = \frac{F}{2m}$

Ans. **[B,D]**

Sol.



$F - f = ma$ …(1)

$Fr = I\alpha$ …(2)

$a = R\alpha$ …(3)

$f = \frac{Ia}{R^2}$

$F - \frac{Ia}{R^2} = ma \Rightarrow F = a.\left(m + \frac{I}{R^2}\right)$

$a = \frac{F}{m + \frac{I}{R^2}}$ a depends on whether cylinder is solid or hollow

For hollow cylinder

$a = \frac{F}{m + \frac{mR^2}{R^2}}$ * $a = \frac{F}{2m}$

Friction force depends on type of cylinder

$f_{max} = \mu mg$

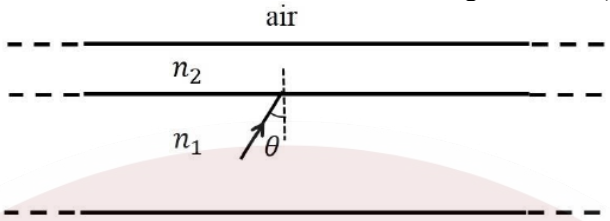
a_{max} for solid cylinder

$a_{max} = \mu mg = \frac{MR^2 a}{2R^2}$

$a_{max} = 2\ \mu g$

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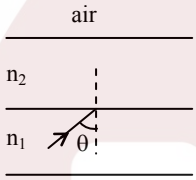
Q.12 A wide slab consisting of two media of refractive indices n_1 and n_2 is placed in air as shown in the figure. A ray of light is incident from medium n_1 to n_2 at an angle θ, where $\sin\theta$ is slightly larger than $1/n_1$. Take refractive index of air as 1. Which of the following statement(s) is (are) correct ?



(A) The light ray enters air if $n_2 = n_1$
(B) The light ray is finally reflected back into the medium of refractive index n_1 if $n_2 < n_1$
(C) The light ray is finally reflected back into the medium of refractive index n_1 if $n_2 > n_1$
(D) The light ray is reflected back into the medium of refractive index n_1 if $n_2 = 1$

Ans. **[B,C,D]**

Sol.

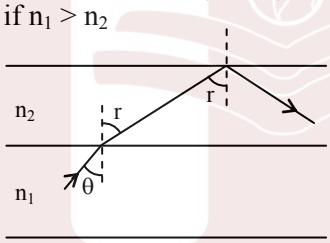


$\sin\theta > \frac{1}{n_1} \Rightarrow \sin\theta = \frac{x}{n_1}$

$\therefore$ θ is greaer than θ_C

T.I.R occurs for ($n_2 = 1$)

if $n_1 > n_2$



$n_1 \sin\theta = n_2 \sin r$

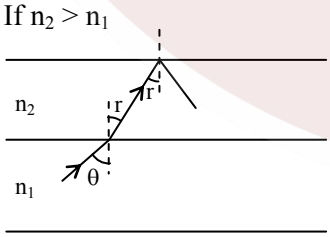
$\sin r = \frac{n_1}{n_2}\sin\theta$

$\sin r > \frac{1}{n_2}$

$\therefore$ r is greater than θ_C for 2nd surface

$\therefore$ TIR occurs at 2nd surface

If $n_2 > n_1$



$n_1 \sin\theta = n_2 \sin r$

$\sin r = \frac{n_1}{n_2}\sin\theta$

$\sin r > \frac{1}{n_2}$

T.I.R occurs at 2nd surface

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Q.13 A particle of mass M = 0.2 kg is initially at rest in the xy-plane at a point (x = –*l*, y = –h), where *l* = 10 m and h = 1 m. The particle is accelerated at time t = 0 with a constant acceleration *a* = 10 m/s^2 along the positive x-direction. Its angular momentum and torque with respect to the origin, in SI units, are represented by $\vec{L}$ and $\vec{\tau}$, respectively. $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along the positive x, y and z-directions, respectively. If $\hat{k} = \hat{i} \times \hat{j}$ then which of the following statement (s) is (are) correct ?

(A) The particle arrives at the point (x = *l*, y = –h) at time t = 2s

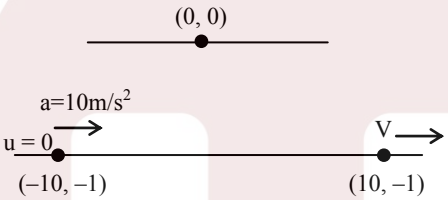
(B) $\vec{\tau} = 2\hat{k}$ when the particle passes through the point (x = *l*, y = –h)

(C) $\vec{L} = 4\hat{k}$ when the particle passes through the point (x = *l*, y = –h)

(D) $\vec{\tau} = \hat{k}$ when the particle passes through the point (x = 0, y = –h)

Ans. **[A,B,C]**

Sol.



$V^2 = u^2 + 2aS$

$V = \sqrt{2 \times 10 \times 20} \Rightarrow V = 20$ m/s

$\vec{\tau}$ on particle will be constant

$= r_\perp F \hat{k} = 1 \times \frac{0.2}{10} \times 10 \hat{k}$

$\vec{\tau} = 2$ N-m $\hat{k}$

$\vec{L}$ at $(10, -1) = r_\perp mv \hat{k}$

$= 1 \times \frac{0.2}{10} \times 20 \hat{k} = 4$ kg m/s^2 $\hat{k}$

time to reach (10, –1)

$S = ut + \frac{1}{2}at^2$

$20 = \frac{1}{2} \times 10t^2 \Rightarrow t = 2$ sec

Q.14 Which of the following statement(s) is (are) correct about the spectrum of hydrogen atom ?

(A) The ratio of the longest wavelength to the shortest wavelength in Balmer series is 9/5

(B) There is an overlap between the wavelength ranges of Balmer and Paschen series

(C) The wavelengths of Lyman series are given by $\left(1 + \frac{1}{m^2}\right)\lambda_0$, where λ_0 is the shortest wavelength of Lyman series and m is an integer

(D) The wavelength of Lyman and Balmer series do not overlap

Ans. **[A,D]**

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Sol. $\frac{1}{\lambda_{min}} = \frac{1}{\lambda_0}\left(\frac{1}{4}-\frac{1}{\infty}\right) \Rightarrow \frac{1}{\lambda_{min}} = \frac{1}{4\lambda_0}$

$\frac{1}{\lambda_{max}} = \frac{1}{\lambda_0}\left(\frac{1}{4}-\frac{1}{9}\right) \Rightarrow \frac{1}{\lambda_{max}} = \frac{5}{4\times 9\lambda_0}$

$\frac{\lambda_{max}}{\lambda_{min}} = \frac{4\times 9\lambda_0}{5\times 4\lambda_0} = \frac{9}{5}$

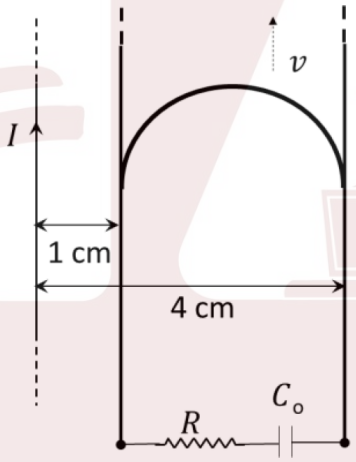
Wavelength rangs of Lyman, Balmer and Paschan series do not overlap.

For Lyman series

$\frac{1}{\lambda} = \frac{1}{\lambda_0}\left(1-\frac{1}{m^2}\right) \Rightarrow \lambda = \left(1-\frac{1}{m^2}\right)^{-1}$

Q.15 A long straight wire carries a current, I = 2 ampere. A semi-circular conducting rod is placed beside it on two conducting parallel rails of negligible resistance. Both the rails are parallel to the wire. The wire, the rod and the rails lie in the same horizontal plane, as shown in the figure. Two ends of the semi-circular rod are at distances 1 cm and 4 cm from the wire. At time t = 0, the rod starts moving on the rails with a speed v = 3.0 m/s (see the figure).

A resistor R = 1.4 Ω and a capacitor C_0 = 5.0 μF are connected in series between the rails. At time t = 0, C_0 is uncharged. Which of the following statement (s) is (are) correct ? [$\mu_0 = 4\pi \times 10^{-7}$ SI units. Take ln 2 = 0.7]



(A) Maximum current through R is 1.2×10^{-6} ampere

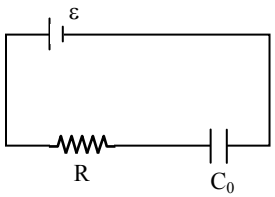
(B) Maximum current through R is 3.8×10^{-6} ampere

(C) Maximum charge on capacitor C_0 is 8.4×10^{-12} coulomb

(D) Maximum charge on capacitor C_0 is 2.4×10^{-12} coulomb

Ans. **[A,C]**

Sol.



$\int d\varepsilon = \int_{x=1}^{x=4}\left(\frac{\mu.i}{2\pi x}\right)(v)\,dx$

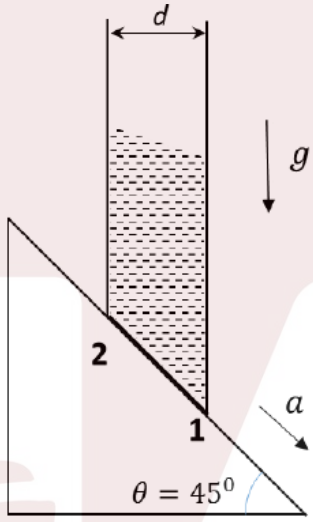
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$\varepsilon = \frac{3\mu_0}{\pi}\ln 4$

$Q_{max} = C_0\varepsilon = 8.4 \times 10^{-12}$ C

$i_{max} = \frac{\varepsilon}{R} = 1.2 \times 10^{-6}$ A

Q.16 A cylindrical tube, with its base as shown in the figure, is filled with water. It is moving down with a constant acceleration *a* along a fixed inclined plane with angle $\theta = 45°$. P_1 and P_2 are pressures at point 1 and 2, respectively, located at the base of the tube. Let $\beta = (P_1 - P_2)/\rho gd)$, where ρ is density of water, d is the inner diameter of the tube and g is the acceleration due to gravity. Which of the following statement (s) is (are) correct ?



(A) $\beta = 0$ when $a = g/\sqrt{2}$

(B) $\beta > 0$ when $a = g/\sqrt{2}$

(C) $\beta = \frac{\sqrt{2}-1}{\sqrt{2}}$ when $a = g/2$

(D) $\beta = \frac{1}{\sqrt{2}}$ when $a = g/2$

Ans. **[A,C]**

Sol. if $a = \frac{g}{\sqrt{2}}$

$\beta = 0$ because tube is falling freely

if $a = \frac{g}{2}$

$P_2 + \frac{\rho g\sqrt{2}\,d}{\sqrt{2}} - \rho\left(\frac{g}{2}\right)\sqrt{2}\,d = P_1$

$P_1 - P_2 = \rho gd\left(1 - \frac{1}{\sqrt{2}}\right)$

$\frac{P_1 - P_2}{\rho gd} = \left(1 - \frac{1}{\sqrt{2}}\right) = \beta$

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SECTION – 4

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numberic keypad in the place-desiginated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme :

  Full Marks : **+4** If ONLY the correct integer is entered.

  Zero Marks : **0** In all other cases.

Q.17 An α-particle (mass 4 amu) and a singly charged sulfur ion (mass 32 amu) are initially at rest. They are accelerated through a potential V and then allowed to pass into a region of uniform magnetic field which is normal to the velocities of the particles. Within this region, the α-particle and the sulphur ion move in circular orbits of radii r_α and r_S, The ratio (r_S/r_α) is ____.

Ans. **[4]**

Sol. $r = \dfrac{mv}{qB}$ $\qquad \dfrac{1}{2}mv^2 = q\Delta V$

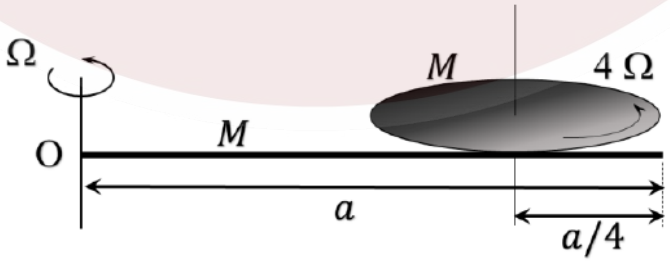
$$mv = \sqrt{2qm\Delta V}$$

$$r = \sqrt{\frac{2m\Delta V}{qB^2}}$$

$$r \propto \sqrt{\frac{m}{q}}$$

$$\frac{r_S}{r_\alpha} = \sqrt{\frac{m_S}{q_S}\frac{q_\alpha}{m_\alpha}} \Rightarrow \frac{r_S}{r_\alpha} = \sqrt{\frac{32}{1}\times\frac{2}{4}} = 4$$

Q.18 A thin rod of mass M and length *a* is free to rotate in horizontal plane about a fixed vertical axis passing through point O. A thin circular disc of mass M and of radius *a*/4 is pivoted on this rod with its center at a distance *a*/4 from the free end so that it can rotate freely about its vertical axis, as shown in the figure. Assume that both the rod and the disc have uniform density and they remain horizontal during the motion. An outside stationary observer finds the rod rotating with an angular velocity Ω and the disc rotating about its vertical axis with angular velocity 4Ω. The total angular momentum of the system about the point O is $\left(\dfrac{Ma^2\Omega}{48}\right)n$. The value of n is _____.



Ans. **[49]**

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Sol. $L = \frac{Ma^2}{3}\Omega + \frac{M(a/4)^2}{2}4\Omega + M\left(\frac{3a}{4}\Omega\right)\left(\frac{3a}{4}\right)$

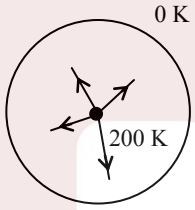
$L = \frac{Ma^2\Omega}{3} + \frac{Ma^2\Omega}{8} + \frac{9Ma^2\Omega}{16}$

$L = \frac{49}{48}Ma^2\Omega$

Q.19 A small object is placed at the center of a large evacuated hollow spherical container. Assume that the container is maintained at 0 K. At time t = 0, the temperature of the object is 200 K. The temperature of the object becomes 100 K at $t = t_1$ and 50 K at $t = t_2$. Assume the object and the container to be ideal black bodies. The heat capacity of the object does not depend on temperature. The ratio (t_2/t_1) is _____.

Ans. **[9]**

Sol.



Power emitted by particle through radiation = σeAT^4

$T_i = 200$ K $\quad T_S = 0$K $\quad e = 1$

$$\frac{dQ}{dt} = -M_S\frac{dT}{dt} = \sigma eAT^4$$

$$-\int_{T_i}^{T_f}\frac{dT}{T^4} = \int_{t_i}^{t_f}\frac{\sigma eA}{M_S}dt$$

$$\frac{\sigma eA}{M_S}(t_f - t_i) = \frac{1}{3}\left(\frac{1}{T_f^3} - \frac{1}{T_i^3}\right)$$

$$\frac{t_0 - 0}{t_2 - 0} = \frac{\left(\frac{1}{100^3} - \frac{1}{200^3}\right)}{\left(\frac{1}{50^3} - \frac{1}{200^3}\right)}$$

$$\frac{t_1}{t_2} = \frac{1}{9} \Rightarrow \frac{t_2}{t_1} = 9$$

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JEE Advanced Exam 2021
(Paper & Solution)

Date : 03 / 10 / 2020

PAPER-1

CHEMISTRY

SECTION – 1

- This section contains **FOUR (04)** questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

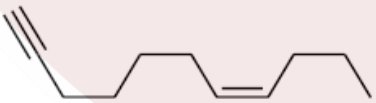
Full Marks	: **+3**	If ONLY the correct option is chosen.
Zero Marks	: **0**	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: **–1**	In all other cases.

Q.1 The major product formed in the following reaction is



(A) 

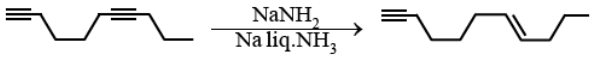
(B) 

(C) 

(D) 

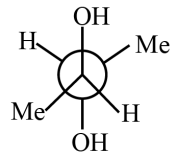
Ans. **[B]**

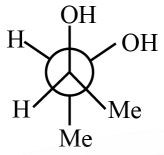
Sol.

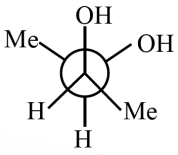


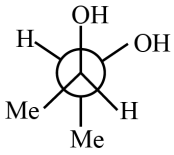
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Q.2 Among the following, the conformation that corresponds to the most stable conformation of meso-butane-2,3-diol is

(A) 

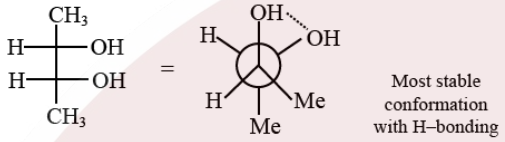
(B) 

(C) 

(D) 

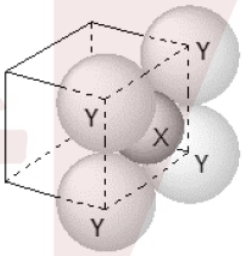
Ans. **[B]**

Sol.



Meso butane –2,3,–diol

Q.3 For the given close packed structure of a salt made of cation X and anion Y shown below (ions of only face are shown for clarity), the packing fraction is approximately (packing fraction = $\frac{\text{packing efficiency}}{100}$)



(A) 0.74 (B) 0.63 (C) 0.52 (D) 0.48

Ans. **[B]**

Sol. Packing fraction (P.F.) = $\frac{1\times\frac{4}{3}\pi r_-^3+3\times\frac{4}{3}\pi r_+^3}{a^3}$

$\frac{r_+}{r_-}$ =0.414 (square planar void), a = 2r_

we get,

P.F. = $\frac{\frac{4}{3}\pi\left(r_-^3+3r_+^3\right)}{8r_-^3}$

$= \left[\frac{\pi}{6}\left(1+3(0.414)^3\right)\right]$

= 0.63

Q.4 The calculated spin only magnetic moments of $[Cr(NH_3)_6]^{3+}$ and $[CuF_6]^{3-}$ in BM, respectively, are (Atomic number of Cr and Cu are 24 and 29, respectively)

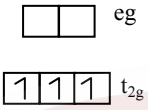
(A) 3.87 and 2.84 (B) 4.90 and 1.73 (C) 3.87 and 1.73 (D) 4.90 and 2.84

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Ans. **[A]**

Sol. $[Cr(NH_3)_6]^{3+}$

$Cr^{3+} \Rightarrow [Ar]3d^3$

In presence of NH_3 ligand 

Number of unpaired electrons = 3

$\mu = \sqrt{n(n+2)}$ B.M.

$\mu = \sqrt{3(3+2)}$ B.M.

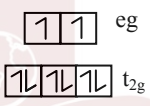
$\mu = \sqrt{15}$ B.M.

$\Rightarrow$ 3.87 B.M.

$[CuF_6]^{3-}$

$Cu^{3+} \Rightarrow [Ar]3d^8$

In presence of F^- Ligand

$Cu^{3+} \Rightarrow$ 

Number of unpaired electrons = 2

$\mu = \sqrt{n(n+2)}$ B.M.

$\mu = \sqrt{2(2+2)} \Rightarrow \sqrt{8}$ B. M.

$\Rightarrow$ 2.84 B.M.

SECTION – 2

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to TWO decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme</u>.

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place ;

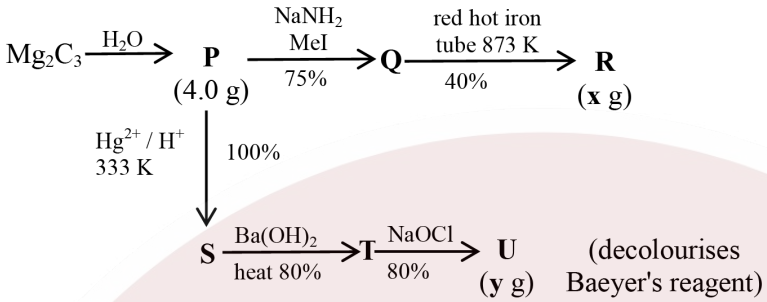
Zero Marks : 0 In all other cases.

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Question Stem for Question Nos. 5 and 6

Question Stem

For the following reaction scheme, percentage yields are given along the arrow :



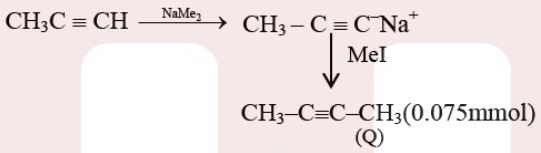
x g and **y** g are mass of **R** and **U**, respectively.

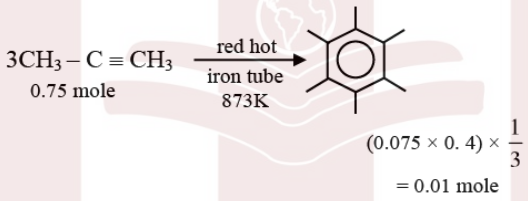
(Use : Molar mass (in g mol^{-1}) of H, C and O as 1, 12 and 16, respectively)

Q.5 The value of **x** is _____.

Ans. **[1.62]**

Sol. $Mg_2C_3 + 4H_2O \rightarrow 2Mg(OH)_2 + CH_3C \equiv CH$





$(0.075 \times 0.4) \times \frac{1}{3}$

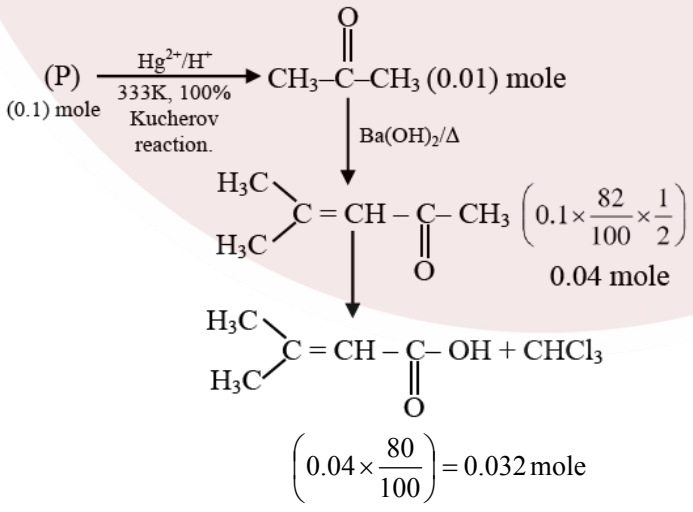
$= 0.01$ mole

The value of x = 162 × 0.01 = 1.62 gm

Q.6 The value of **y** is _____.

Ans. **[3.2]**

Sol.



$\left(0.1 \times \frac{82}{100} \times \frac{1}{2}\right)$

$\left(0.04 \times \frac{80}{100}\right) = 0.032$ mole

60 + 32 + 8 = 100

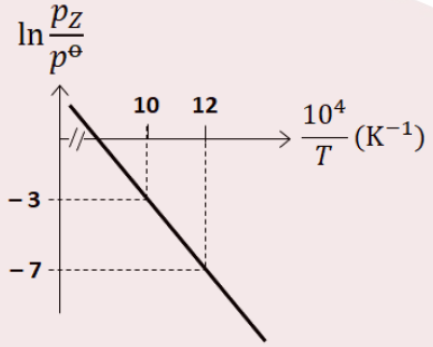
The value of Y = 0.032 × 100 = 3.2

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Question Stem for Question Nos. 7 & 8

Question Stem

For the reaction $X(s) \rightleftharpoons Y(s) + Z(g)$, the plot of $\ln \frac{p_z}{p^\theta}$ versus $\frac{10^4}{T}$ is given below (in solid line), where p_z is the pressure (in bar) of the gas Z at temperature T and $p^\theta = 1$ bar.



(Given, $\frac{d(\ln K)}{d\left(\frac{1}{T}\right)} = -\frac{\Delta H^\theta}{R}$, where the equilibrium constant, $K = \frac{p_z}{p^\theta}$ and the gas constant, R = 8.314 J K^{-1} mol^{-1}**)**

Q.7 The value of standard enthalpy, ΔH^θ (in kJ mol^{-1}) for the given reaction is _____.

Ans. **[166.28]**

Sol. $\Delta G^\circ = -RT \ln\left(\frac{P}{1}\right) = \Delta H^\circ - T\Delta S^\circ$

$\ln\left(\frac{P}{1}\right) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$

$\text{Slope} = -\frac{\Delta H^\circ}{RT} = 10^4 \times \left(-\frac{4}{2}\right)$

$\Rightarrow \Delta H^\circ = 2 \times 10^4 \times R$

$= 166.28$ kJ/mole

Q.8 The value of ΔS^θ (in J K^{-1} mol^{-1}) for the given reaction, at 1000 K is_____.

Ans. **[141.33 or 141.34]**

Sol. From the plot when, $\frac{10^4}{T} = 10 \Rightarrow T = 1000$ K

$\ln\left(\frac{P_2}{1}\right) = -3$

Substituting in equation :

$\ln\left(\frac{P_2}{1}\right) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$

We get,

$-3 = -\frac{2 \times 10^4 \times R}{R \times 1000} + \frac{\Delta S^\circ}{R}$

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$\Rightarrow \Delta S° = 17R$
$\Rightarrow \Delta S° = 17 \times 8.314$ J/K-mol
$\Rightarrow \Delta S° = 141.34$ J/K-mol

Question Steam for Question Nos. 9 & 10

Question Steam

The boiling point of water in a 0.1 molal silver nitrate solution (Solution **A**) is **x** °C. To this solution **A**, an equal volume of 0.1 molal aqueous barium chloride solution is added to make a new solution **B**. The difference in the boiling points of water in the two solutions **A** and **B** is **y** $\times 10^{-2}$ °C.

(Assume : Densities of the solution **A** and **B** are the same as that of water and the soluble salts dissociate completely.

Use : Molal elevation constant (Ebullioscopic Constant), $K_b = 0.5$ K kg mol^{-1} ; Boiling point of pure water as 100° C.)

Q.9 The value of x is _____.

Ans. **[100.10]**

Sol. $AgNO_3\,(aq) \longrightarrow Ag^+\,(aq) + NO_3^-\,(aq)$

0.1 m 0.1 m

$\Delta T_b = 0.2 \times 0.5$

$= 0.1°C = 0.1$ K

Boiling point of solution = 100.1°C

= X

Q.10 The value of |y| is _____.

Ans. **[2.50]**

Sol. $AgNO_3\,(aq) \longrightarrow Ag^+(aq) + NO_3^-\,(aq)$

0.05 m 0.05 m 0.05 m

$BaCl_2(aq) \longrightarrow Ba^{2+}\,(aq) + 2Cl^-\,(aq)$

0.05 m 0.05 m 0.1 m

Ag^+ and Cl^- combine to form AgCl precipitate

$Ag^+\,(aq) + Cl^-\,(aq) \rightarrow AgCl(s)$

t = 0	0.05 m	0.1 m
t = ∞	0	0.05 m

In final solution total concentration of all ions :

$[Cl^-] + [NO_3^-] + [Ba^{2+}] = 0.05 + 0.05 + 0.05$

= 0.15 m

$\Delta T_b = 0.5 \times 0.15$

= 0.075 °C

B.P. of solution 'B' = 100.075 °C

B.P. of solution 'A' = 100.1°C

|y| = 100.1 – 100.075

$= 0.025 = 2.5 \times 10^{-2}$

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SECTION – 3

- This section contains **SIX (06)** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	: **+4**	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: **+3**	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: **+2**	If three or more options are correct but ONLY two options are chosen, both of which are correct.
Partial Marks	: **+1**	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: **0**	If unanswered.
Negative Marks	: **–2**	In all other cases.

- For example, in a question (A), (B) and C) and (D) are the ONLY three options corresponding to correct answer, then

  choosing ONLY (A), (B) an (D) will get +4 marks ;

  choosing ONLY (A) and (B) will get +2 marks;

  choosing ONLY (A) and (D) will get +2 marks ;

  choosing ONLY (B) and (D) will get +2 marks;

  choosing ONLY (A) will get +1 mark;

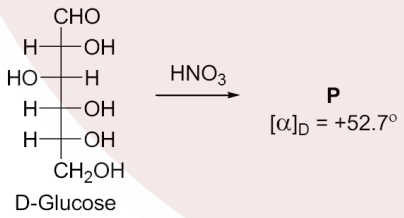
  choosing ONLY (B) will get +1 mark;

  choosing ONLY (D) will get +1 mark;

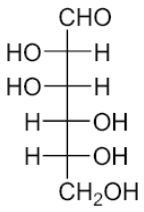
  choosing no option (s) (i.e. the question is unanswered) will get 0 marks and

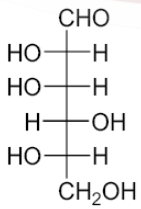
  choosing any other option (s) will get –2 marks.

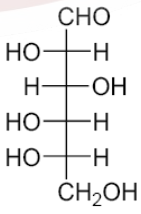
Q.11 Given :

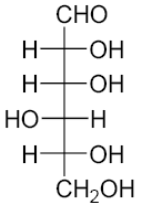


The compound(s), which on reaction with HNO_3 will give the product having degree of rotation, $[\alpha]_D$ = – 52.7° is (are)

(A) 

(B) 

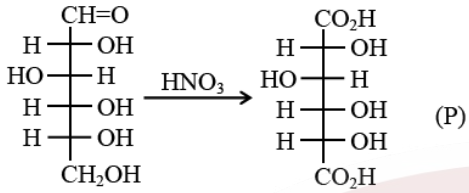
(C) 

(D) 

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Ans. **[C,D]**

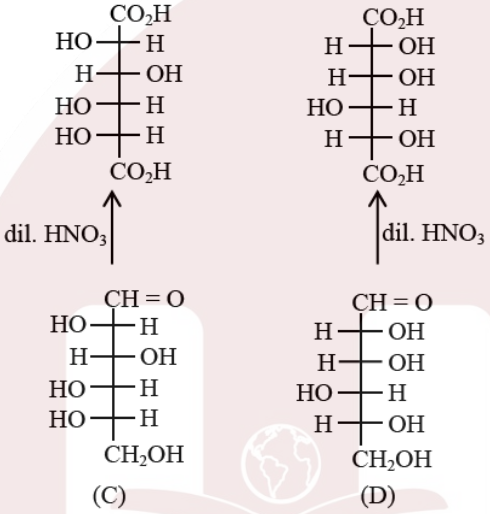
Sol.



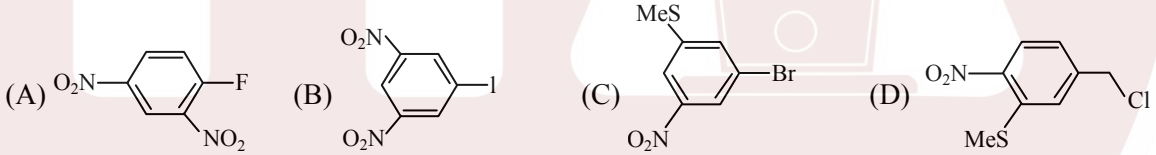
(P)

$[\alpha]_D = 52.7°$

The enantiomer of P has rotation – 52.7° is as follows

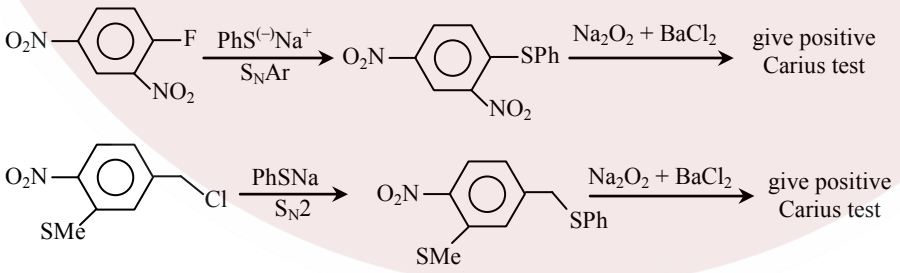


Q.12 The reaction of **Q** with PhSNa yields an organic compound (major product) that gives positive Carius test on treatment with Na_2O_2 followed by addition of $BaCl_2$. The correct options (s) for **Q** is (are)



Ans. **[A,D]**

Sol.



Q.13 The correct statement (s) related to colloids is (are)

(A) The process of precipitating colloidal sol by an electrolyte is called peptization.

(B) Colloidal solution freezes at higher temperature than the true solution at the same concentration.

(C) Surfactants form micelle above critical micelle concentration (CMC). CMC depends on temperatue.

(D) Micelles are marcomolecular colloids.

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Ans. **[B,C]**

Sol. (A) Process of precipitating colloidal solution is called coagulation. Hence false.

(B) For colloidal solutions concentration is very small due to very large molar mass and hence their colligative properties are very small as compared to true solutions

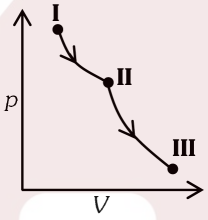
$\therefore \Delta T_f$ is lesser for colloidal solution. Hence true.

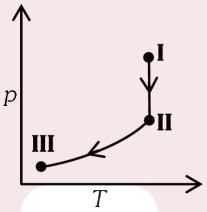
(C) At CMC surfactant form micelles. Hence true

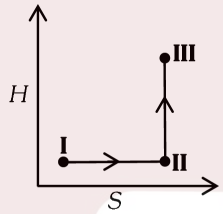
(D) Micelles and macromolecular colloids are two different types of colloids. Hence false.

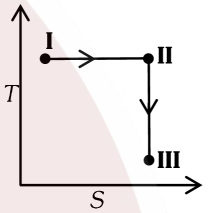
Q.14 An ideal gas undergoes a reversible isothermal expansion from state I to state II followed by a reversible adiabatic expansion from state **II** to state **III**. The correct plot (s) representing the changes from state **I** to state **III** is (are)

(P : pressure, V : volume, T : temperature, H : enthalpy, S : entropy)

(A) 

(B) 

(C) 

(D) 

Ans. **[A,B,D]**

Sol. From state I to II (Reversible isothermal expansion)

$\Rightarrow$ P decreases, V increases, T constant

H constant & S increases.

From state II to III (Reversible adiabatic expansion)

$\Rightarrow$ P decreases, V increases, T decreases

H decreases, S constant

$\therefore$ Plots (A), (B), (D) are correct while (C) is wrong as from II to III, H is decreasing.

Q.15 The correct statement(s) related to the metal extraction processes is (are)

(A) A mixture of Pbs and PbO undergoes self-reduction to produce Pb and SO_2.

(B) In the extraction process of copper from copper pyrites, silica is added to produce copper silicate.

(C) Partial oxidation of sulphide ore of copper by roasting, followed by self-reduction produces blister copper.

(D) In cyanide process, zinc powder is utilized to precipitate gold from Na $[Au(CN)_2]$.

Ans. **[A,C,D]**

Sol. (A) $PbS + 2PbO \rightarrow 3Pb + SO_2$ (self reduction)

(B) Silica is added to remove impurity of Fe in the form of slag $FeSiO_3$

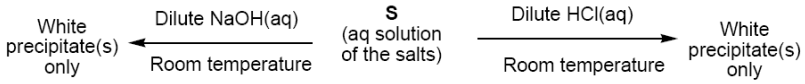
(C) $CuFeS_2$ ore is partially oxidized first by roasting and then self reduction of Cu takes place to produce blister copper.

(D) $4\,Na\,[\overset{+1}{Au}\,(CN)_2] + 2\,Zn \longrightarrow 2\,Na_2[\overset{+2}{Zn}(CN)_4] + 4\,\overset{0}{Au}$

Reducing Agent

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Q.16 A mixture of two salts is used to prepare a solution S, which gives the following results :



The correct option (s) for the salt mixture is (are)

(A) $Pb(NO_3)_2$ and $Zn(NO_3)_2$ (B) $Pb(NO_3)_2$ and $Bi(NO_3)_3$

(C) $AgNO_3$ and $Bi(NO_3)_3$ (D) $Pb(NO_3)_2$ and $Hg(NO_3)_2$

Ans. **[A,B]**

Sol. $Pb(NO_3)_2 \xrightarrow{\text{dil. HCl}} \underset{\text{White PPt.}}{PbCl_2 \downarrow}$

$Bi(NO_3)_3 \xrightarrow{\text{dil. HCl}} \underset{\text{WaterSoluble}}{BiCl_3}$

$Hg(NO_3)_2 \xrightarrow{\text{dil. HCl}} \underset{\text{Water Soluble}}{HgCl_2}$

$AgNO_3 \xrightarrow{\text{dil. HCl}} \underset{\text{WhitePPt.}}{AgCl \downarrow}$

$Zn(NO_3)_2 \xrightarrow{\text{dil. HCl}} \underset{\text{Water Soluble}}{ZnCl_2}$

$Pb(NO_3)_2 \xrightarrow{\text{NaOH(dil.)}} \underset{\text{White PPt.}}{Pb(OH)_2 \downarrow}$

$Zn(NO_3)_2 \xrightarrow{\text{NaOH(dil.)}} \underset{\text{White PPt.}}{Zn(OH)_2 \downarrow}$

$Bi(NO_3)_3 \xrightarrow{\text{NaOH(dil.)}} \underset{\text{White PPt.}}{Bi(OH)_3 \downarrow}$

$AgNO_3 \xrightarrow{\text{NaOH(dil.)}} \underset{\text{Brown PPt.}}{Ag_2O}$

$Hg(NO_3)_2 \xrightarrow{\text{NaOH(dil.)}} \underset{\text{Yellow PPt.}}{HgO \downarrow}$

SECTION – 4

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each quesiton, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numberic keypad in the place-desiginated to enter the answer.
- Answer to each question will be evaluated <u>according to the following marking scheme</u> :

  Full Marks : **+4** If ONLY the correct integer is entered.

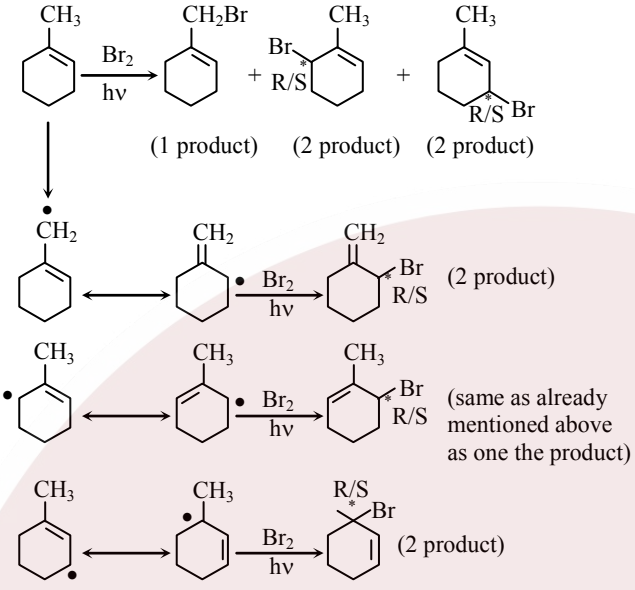
  Zero Marks : **0** In all other cases.

Q.17 The maximum number of possible isomers (including stereoisomers) which may be formed on mono-bromination of 1-methylcyclohex-1-ene using Br_2 and UV light is ____.

Ans. **[9]**

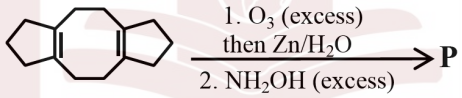
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Sol.



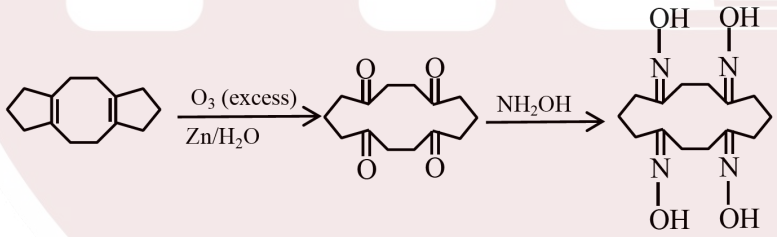
Total 9 product

Q.18 In the reaction given below, the total number of atoms having sp^2 hybridization in the major product **P** is____.



Ans. **[8]**

Sol.



total 8 atoms are sp^2 hybridised

Q.19 The total number of possible isomers for $[Pt(NH_3)_4Cl_2]Br_2$ is ____.

Ans. **[6]**

Sol. **Isomers**

(I) $[Pt(NH_3)_4Cl_2]Br_2 \Rightarrow$ G.I. = 2

(II) $[Pt(NH_3)_4Br2]Cl_2 \Rightarrow$ G.I. = 2

(III) $[Pt(NH_3)_4BrCl]Br.Cl \Rightarrow$ G.I. = 2

I, II, III are ionisation isomers of each other, each having 2 geometrical isomers.

Total possible isomers will be 6

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JEE Advanced Exam 2021
(Paper & Solution)

PAPER-1

MATHEMATICS

SECTION – 1

- This section contains **FOUR (04)** questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	: **+3**	If ONLY the correct option is chosen.
Zero Marks	: **0**	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: **–1**	In all other cases.

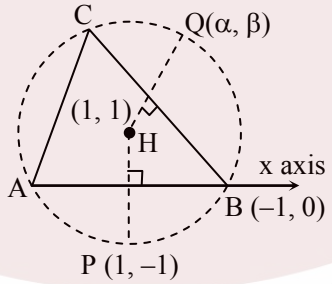
Q.1 Consider a triangle Δ whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocenter of Δ is (1, 1), then the equation of the circle passing through the vertices of the triangle Δ is

(A) $x^2 + y^2 - 3x + y = 0$ (B) $x^2 + y^2 + x + 3y = 0$

(C) $x^2 + y^2 + 2y - 1 = 0$ (D) $x^2 + y^2 + x + y = 0$

Ans. **[B]**

Sol. The required circle must pass through (–1, 0). Image of the orthocenter H(1, 1) w.r.t the sides of Δ must lie on the circumcircle.



Hence P(1, –1) and Q(image of H in $x + y + 1 = 0$) lies on the required circle

so $\frac{\alpha - 1}{1} = \frac{\beta - 1}{1} = \frac{-2(1+1+1)}{2}$

$\alpha = -2, \beta = -2$

option (B) satisfies their three points

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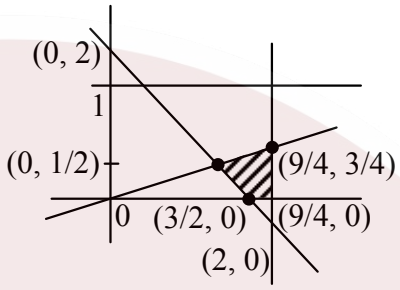
Q.2 The area of the region $\{(x, y) : 0 \le x \le 9/4;\ 0 \le y \le 1;\ x \ge 3y;\ x + y \ge 2\}$ is

(A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

Ans. **[A]**

Sol. Required Area

$$= \frac{1}{2}\begin{vmatrix} 2 & 0 \\ 9/4 & 0 \\ 9/4 & 3/4 \\ 3/2 & 1/2 \\ 2 & 0 \end{vmatrix}$$



$$= \frac{1}{2}\left(0 + \frac{27}{16} - 0 + \frac{9}{8} - \frac{9}{8} + 0 - 1\right)$$

$$= \frac{1}{2} \times \frac{27 - 16}{16} = \frac{11}{32}$$

Q.3 Consider three sets $E_1 = \{1, 2, 3\}$, $F_1 = \{1, 3, 4\}$ and $G_1 = \{2, 3, 4, 5\}$. Two elements are chosen at random, without replacement, form the set E_1, and let S_1 denote the set of these chosen elements. Let $E_2 = E_1 - S_1$ and $F_2 = F_1 \cup S_1$. Now two elements are chosen at random, without replacement, from the set F_2 and let S_2 denote the set of these chosen elements.

Let $G_2 = G_1 \cup S_2$. Finally, two elements are chosen at random, without replacement, from the set G_2 and let S_3 denote the set of these chosen elements.

Let $E_3 = E_2 \cup S_3$. Given that $E_1 = E_3$, let p be the conditional probability of the event $S_1 = \{1, 2\}$. Then the value of p is

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{1}{2}$ (D) $\frac{2}{5}$

Ans. **[A]**

Sol. Let us define the events

$A \equiv E_1 = E_3$	$P\left(\frac{A}{A_1}\right) = \frac{3}{6} \times \frac{1}{10}$
$A_1 = \{1, 2\}$	$P\left(\frac{A}{A_2}\right) = \frac{2}{3} \times \frac{1}{10}$
$A_2 = \{1, 3\}$	$P\left(\frac{A}{A_3}\right) = \frac{3}{6} \times \frac{1}{6} + \frac{3}{6} \times \frac{1}{10}$
$A_3 = \{2, 3\}$	$P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}$

Required probability

$$P = P\left(\frac{A_1}{A}\right) = \frac{P\left(\frac{A}{A_1}\right).P(A_1)}{P\left(\frac{A}{A_1}\right)P(A_1) + P\left(\frac{A}{A_2}\right)P(A_2) + P\left(\frac{A}{A_3}\right)P(A_3)}$$

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$$= \frac{\frac{3}{6}\times\frac{1}{10}}{\frac{3}{6}\times\frac{1}{10}+\frac{2}{3}\times\frac{1}{10}+\frac{3}{6}\times\frac{1}{6}+\frac{3}{6}\times\frac{1}{10}}$$

$$= \frac{3}{3+4+\frac{3}{6}\times10\times3} = \frac{3}{3+4+5+3} = \frac{1}{5}$$

Q.4 Let $\theta_1, \theta_2, ... \theta_{10}$ be positive valued angles (in radian) such that $\theta_1 + \theta_2 + ... + \theta_{10} = 2\pi$. Define the complex numbers $z_1 = e^{i\theta_1}$, $z_k = z_{k-1}\ e^{i\theta_k}$ for k = 2, 3, ...,10, where $i = \sqrt{-1}$. Consider the statements P and Q given below :

P : $|z_2 - z_1| + |z_3 - z_2| + ... + |z_{10} - z_9| + |z_1 - z_{10}| \le 2\pi$

Q : $|\ z_2^2 - z_1^2\ | + |\ z_3^2 - z_2^2\ | + ... + |\ z_{10}^2 - z_9^2\ | + |\ z_1^2 - z_{10}^2\ | \le 4\pi$

Then,

(A) P is **True** and Q is **False**

(B) Q is **True** and P is **False**

(C) both P and Q are **True**

(D) both P and Q are **False**

Ans. **[C]**

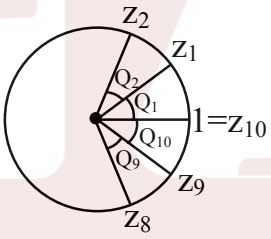
Sol. $|z_1| = 1$

$|z_K| = |z_{K-1}|$ for $K \ge 2$

so $|z_1| = |z_2| = ... = |z_{10}| = 1$

since $\theta_1 + \theta_2 + ... + \theta_{10} = 2\pi$

so $Z_{10} = 1$



Now, since $|z_1 - z_2| \le r.\ \theta_2$

$$|z_1 - z_2| \le \theta_2 \quad \text{here } r = 1$$

so $|z_2 - z_1| + |z_4 - z_3| + ...+ |z_{10} - z_9| + |z_1 - z_{10}| \le \theta_2 + \theta_3 + \theta_4 + ... + \theta_9 + \theta_{10}$

so $|z_2 - z_1| + |z_4 - z_3| + ...+ |z_{10} - z_9| + |z_1 - z_{10}| \le 2\pi$

Hence, statement P is true

Now since $|\ z_1^2 - z_2^2\ | = |z_1 - z_2|\ |z_1 + z_2|$

$$|\ z_1^2 - z_2^2\ | \le |z_1 - z_2|\ (|z_1 + z_2|)$$

so $|\ z_1^2 - z_2^2\ | \le 2|z_1 - z_2|$ as $|z_1| = |z_2| = 1$

Hence again by using $|z_1 - z_2| \le \theta_2$

$|\ z_2^2 - z_1^2\ | + |\ z_2^2 - z_3^2\ | + ...+ |\ z_{10}^2 - z_9^2\ | + |\ z_1^2 - z_{10}^2\ | \le 2.2\pi$

Hence statement Q is also correct.

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SECTION – 2

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme.

Full Marks : +2 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 5 & 6

Question Stem

Three numbers are chosen at random, one after another with replacement, from the set S = {1, 2, 3, ..., 100}. Let p_1 be the probability that the maximum of chosen numbers is at least 81 and p_2 be the probability that the minimum of chosen numbers is at most 40.

Q.5 The value of $\frac{625}{4}p_1$ is .....

Ans. **[76.25]**

Sol. Probability that a number ≥ 80 is chosen = $\frac{20}{100}$

Hence $p_1 = 1 - \left(\frac{80}{100}\right)^3 = 1 - \left(\frac{4}{5}\right)^3 = 1 - \frac{64}{125} = \frac{61}{125}$

Similarly probability that the chosen number is ≤ 40 = $\frac{40}{100}$

Hence $p_2 = 1 - \left(\frac{60}{100}\right)^3 = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{27}{125} = \frac{98}{125}$

So, Now

$\frac{625}{4}p_1 = \frac{625}{4} \times \frac{61}{125} = \frac{305}{4} = 76.25$

Q.6 The value of $\frac{125}{4}p_2$ is .....

Ans. **[74.50]**

Sol. Probability that a number ≥ 80 is chosen = $\frac{20}{100}$

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Hence $p_1 = 1 - \left(\frac{80}{100}\right)^3 = 1 - \left(\frac{4}{5}\right)^3 = 1 - \frac{64}{125} = \frac{61}{125}$

Similarly probability that the chosen number

is $\le 40 = \frac{40}{100}$

Hence $p_2 = 1 - \left(\frac{60}{100}\right)^3 = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{27}{125} = \frac{98}{125}$

So, Now

$\frac{125}{4} p_2 = \frac{125}{4} \times \frac{98}{125} = 24.50$

Question Stem for Question Nos. 7 & 8

Question Stem

Let α, β and γ be real numbers such the system of linear equations

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

is consistent. Let |M| represent the determinant of the matrix

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equation is consistent, and D be the square of the distance of the point (0, 1, 0) form the plane P.

Q.7 The value of |M| is .....

Ans. **[1]**

Sol. $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 3 \begin{vmatrix} 1 & 2 & 1 \\ 4 & 5 & 2 \\ 7 & 8 & 3 \end{vmatrix}$

$= 3[15 - 16 - 2(12 - 14) + (32 - 35)]$

$= 3[-1 + 4 - 3] = 0$

So, $\Delta_1 = \Delta_2 = \Delta_3 = 0$ as system is consistent

$$\Delta_1 = \begin{vmatrix} \alpha & 2 & 3 \\ \beta & 5 & 6 \\ \gamma - 1 & 8 & 9 \end{vmatrix} = 3 \begin{vmatrix} \alpha & 2 & 1 \\ \beta & 5 & 2 \\ \gamma - 1 & 8 & 3 \end{vmatrix} = 0$$

$= 3[\alpha(15 - 16) - 2(3\beta - 2\gamma + 2) + (8\beta - 5\gamma + 5)] = 0$

$= 3(-\alpha - 6\beta + 4\gamma - 4 + 8\beta - 5\gamma + 5) = 0$

$\Rightarrow -\alpha + 2\beta - \gamma + 1 = 0$

$\Rightarrow \alpha - 2\beta + \gamma - 1 = 0$

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So Now,

$$D = \left|\frac{0-2+8-1}{\sqrt{1+4+1}}\right|^2$$

$$D = \frac{9}{6} = \frac{3}{2} = 1.5$$

Now $|M| = \alpha(1) - 2(\beta) + \gamma(1)$

$= \alpha - 2\beta + \gamma = 1$

Q.8 The value of D is .....

Ans. **[1.5]**

Sol. $$\Delta = \begin{vmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{vmatrix} = 3\begin{vmatrix}1 & 2 & 1\\ 4 & 5 & 2\\ 7 & 8 & 3\end{vmatrix}$$

$= 3[15 - 16 - 2(12 - 14) + (32 - 35)]$

$= 3[-1 + 4 - 3] = 0$

So, $\Delta_1 = \Delta_2 = \Delta_3 = 0$ as system is consistent

$$\Delta_1 = \begin{vmatrix}\alpha & 2 & 3\\ \beta & 5 & 6\\ \gamma-1 & 8 & 9\end{vmatrix} = 3\begin{vmatrix}\alpha & 2 & 1\\ \beta & 5 & 2\\ \gamma-1 & 8 & 3\end{vmatrix} = 0$$

$= 3[\alpha(15 - 16) - 2(3\beta - 2\gamma + 2) + (8\beta - 5\gamma + 5)] = 0$

$= 3(-\alpha - 6\beta + 4\gamma - 4 + 8\beta - 5\gamma + 5) = 0$

$\Rightarrow -\alpha + 2\beta - \gamma + 1 = 0$

$\Rightarrow \alpha - 2\beta + \gamma - 1 = 0$

So Now,

$$D = \left|\frac{0-2+8-1}{\sqrt{1+4+1}}\right|^2$$

$$D = \frac{9}{6} = \frac{3}{2} = 1.5$$

Question Stem for Question Nos. 9 & 10

Question Stem

Consider the lines L_1 and L_2 defined by

$L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant λ, let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2. The line $y = 2x + 1$ meets C at two points R and S, where the distance between R and S is $\sqrt{270}$.

Let the perpendicular bisector of RS meet C at two distinct points R′ and S′. Let D be the **square** of the distance between R′ and S′.

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Q.9 The value of λ^2 is .....

Ans. **[9]**

Sol. $L_1 \equiv y = -\sqrt{2}x + 1$

$L_2 \equiv y = \sqrt{2}x + 1$

$L \equiv y = 2x + 1$

Let P(x, y)

So, $\left|\frac{x\sqrt{2}+y-1}{\sqrt{2+1}}\right| \left|\frac{x\sqrt{2}-y+1}{\sqrt{2+1}}\right| = \lambda^2$

$\frac{|2x^2-(y-1)^2|}{3} = \lambda^2$ ...(1)

Since line L exist it at R & S

So solve it with L

$\left|\frac{2x^2-4x^2}{3}\right| = \lambda^2$

$2x^2 = 3\lambda^2$

$x = \pm\sqrt{\frac{3}{2}}\lambda$

Now, RS = $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2} = \sqrt{270}$

$= \sqrt{(x_1-x_2)^2+4(x_1-x_2)^2} = \sqrt{270}$

$= \sqrt{5}\,|x_1-x_2| = \sqrt{270}$

$= \sqrt{5}\,.\,2\sqrt{\frac{3}{2}}\lambda = \sqrt{270}$

$\lambda = \sqrt{\frac{270}{3.10}} = \sqrt{9} = 3$

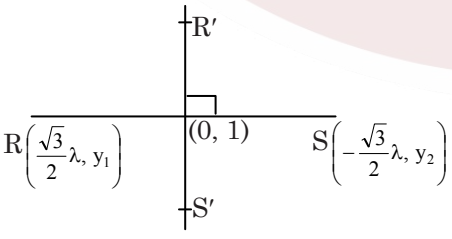
So, $\lambda^2 = 9$

Q.10 The value of D is .....

Ans. **[77.14]**

Sol. Now R′S′ = $y - 1 = -\frac{1}{2}x$

So, solve it with (1)



$\left|2x^2-\frac{1}{4}x^2\right| = 3.9$

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$7x^2 = 12.9$

$x = \pm 6\sqrt{\frac{3}{7}}$

Now, $D = (x_1 - x_2)^2 + (y_1 - y_2)^2$

$= (x_1 - x_2)^2 + \frac{1}{4}(x_1 - x_2)^2$

$= \frac{5}{4}\left(2.6\sqrt{\frac{3}{7}}\right)^2$

$= \frac{5 \times 36 \times 3}{7}$

$= \frac{540}{7} = 77.14$

SECTION – 3

- This section contains **SIX (06)** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	: **+4**	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: **+3**	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: **+2**	If three or more options are correct but ONLY two options are chosen, both of which are correct.
Partial Marks	: **+1**	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: **0**	If unanswered;
Negative Marks	: **–2**	In all other cases.

- For example, in a question (A), (B) and C) and (D) are the ONLY three options corresponding to correct answer, then

  choosing ONLY (A), (B) an (D) will get +4 marks ;

  choosing ONLY (A) and (B) will get +2 marks;

  choosing ONLY (A) and (D) will get +2 marks ;

  choosing ONLY (B) and (D) will get +2 marks;

  choosing ONLY (A) will get +1 mark;

  choosing ONLY (B) will get +1 mark;

  choosing ONLY (D) will get +1 mark;

  choosing no option (s) (i.e. the question is unanswered) will get 0 marks and

  choosing any other option (s) will get –2 marks.

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Q.11 For any 3 × 3 matrix M, let |M| denote the determinant of M. Let

$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and $F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$.

If Q is a nonsingular matrix of order 3 × 3, then which of the following statements is (are) **TRUE**?

(A) $F = PEP$ and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

(B) $|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$

(C) $|(EF)^3| > |EF|^2$

(D) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

Ans. **[A,B,D]**

Sol. (A) $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Now, find PEP (check yourself) it is $PEP = P^2$

(B) $|EQ + PFQ^{-1}| = |EQ + EPQ^{-1}| = |E| \times |Q + PQ^{-1}|$

as $F = PEP \Rightarrow PF = P^2EP = EP$

$|EQ + PFQ^{-1}| = |E| \times |Q + Q^{-1}| = 0$ as $|E| = 0$

Now, $|EQ| + |PFQ^{-1}| = |E|\,|Q| + |EP|$

$= |E|\,(|Q| + |E|\,|P|) = 0$ as $|E| = 0$

So RHS = LHS

(C) $|(EF)^3| > |EF|^2$ wrong as $|E| = 0$

(D) $P^{-1}EP + F = PEP + F = F + F = 2F$

and $P^{-1}FP + E = PFP + E = E + E = 2E$

as $PFP = PPEPP$

$= P^2EP^2 = E$

Now, check tr of 2F & tr 2E they are same.

Q.12 Let $f : R \to R$ be defined by

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

Then which of the following statements is (are) **TRUE**?

(A) f is decreasing in the interval (–2, –1)

(B) f is increasing in the interval (1, 2)

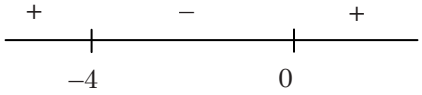
(C) f is onto

(D) Range of f is $\left[-\frac{3}{2}, 2\right]$

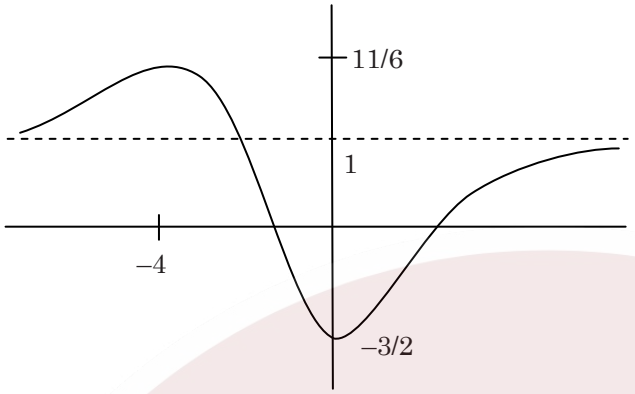
Ans. **[A, B]**

Sol. $F' = \dfrac{(2x-3)(x^2+2x+6) - (x^2-3x-6)(2x+2)}{(x^2+2x+4)^2}$

$= \dfrac{5x^2+20x}{(x^2+2x+4)^2} = \dfrac{5x(x+4)}{(x^2+2x+4)^2}$



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$f_{(max)}(x) = f(-4) = \frac{11}{6}$

$f_{min}(x) = f(0) = -\frac{6}{4} = -\frac{3}{2}$

$\lim\limits_{x \to \pm\infty} f(x) = 1$

Range $= \left[-\frac{3}{2}, \frac{11}{6}\right]$

Q.13 Let E, F and G be three events having probabilities $P(E) = \frac{1}{8}$, $P(F) = \frac{1}{6}$ and $P(G) = \frac{1}{4}$, and let $P(E \cap F \cap G) = \frac{1}{10}$. For any event H, if H^C denotes its complement, then which of the following statements is (are) **TRUE**?

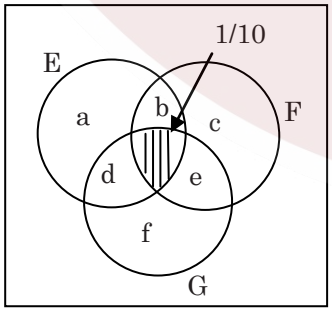
(A) $P(E \cap F \cap G^C) \le \frac{1}{40}$ (B) $P(E^C \cap F \cap G) \le \frac{1}{15}$

(C) $P(E \cup F \cup G) \le \frac{13}{24}$ (D) $P(E^C \cap F^C \cap G^C) \le \frac{5}{12}$

Ans. **[A,B,C]**

Sol. $P(E) = a + b + c + d + e + \frac{1}{10} = \frac{1}{8}$

$a + b + d = -\frac{1}{10} + \frac{1}{8} = \frac{-8+10}{80} = \frac{1}{40}$ ...(1)



$P(F) = b + c + e + \frac{1}{10} = \frac{1}{6}$

$b + c + e = \frac{1}{6} - \frac{1}{10} = \frac{10-6}{60} = \frac{1}{15}$ ...(2)

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$P(G) = d + e + f + \frac{1}{10} = \frac{1}{4}$

$d + e + f = \frac{1}{4} - \frac{1}{10} = \frac{5-2}{20} = \frac{3}{20}$ ...(3)

Now, (A) $P(E \cap F \cap G^C) = b < b + a + d < \frac{1}{4}$ ...(A)

(B) $P(E^C \cap F \cap G) = e < \frac{1}{15} - h - e < \frac{1}{15}$ ...(B)

(C) $P(E \cup F \cup G) \le \frac{1}{8} + \frac{1}{6} + \frac{1}{4}$

$= \frac{3+4+6}{24} = \frac{13}{24}$ ...(C)

(D) $P(E^C \cap F^C \cap G^C) = 1 - P(E \cup F \cup G) \ge 1 - \frac{13}{24} \ge \frac{1}{2}$

Q.14 For any 3 × 3 matrix M, let |M| denote the determinant of M. Let I be the 3 × 3 identity matrix. Let E and F be two 3 × 3 matrices such that (I – EF) is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) **TRUE**?

(A) |FE| = |I – FE| |FGE| (B) (I – FE) (I + FGE) = I

(C) EFG = GEF (D) (I – FE) (I – FGE) = I

Ans. **[A,B,C]**

Sol. $G = (I - EF)^{-1}$

Since

$(I - EF) \cdot G = I = G(I - EF)$

$G - EFG = G - GEF$

$EFG = GEF$ ___(C)

Now Since $FGE = F(I - EF)^{-1} E$

$= (E^{-1}(I - EF) F^{-1})^{-1}$

$FGE = (E^{-1} F^{-1} - I)^{-1}$

So, $|FGE| |E^{-1} F^{-1} - I| = I$

$|FGE| \, | E^{-1} F^{-1} - I| \, |F E| = |FE|$

$|FGE| \; |I - FE| = |FE|$ ____(A)

Now (I – FE) (I + FGE)

= I + FGE – FE – FEFGE

= I + FGE – FE – FGEFE

= I + FGE – FE – F(G – I) E as (I – EF) G = I

= I + FGE – FE – FGE + EE

= I

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Q.15 For any positive integer n, let $S_n : (0, \infty) \to R$ be defined by

$$S_n(x) = \sum_{k=1}^{n} \cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right),$$

where for any $x \in R$, $\cot^{-1}(x) \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) **TRUE**?

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$, for all $x \geq 0$

(B) $\lim_{n\to\infty} \cot(S_n(x)) = x$, for all $x \geq 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

Ans. **[A,B]**

Sol. $S_n(x) = \sum_{k=1}^{n} \cot^{-1}\frac{1+(k+1)x.kx}{(k+1)x-kx}$

$= \sum_{k=1}^{n} \cot^{-1}(k+1)x - \cot^{-1}kx$

$= \cot^{-1}(n+1)x - \cot^{-1}k$

$= \cot^{-1}\frac{1+(n+1)x^2}{(n+1)x-x} = \cot^{-1}\frac{1+(n+1)x^2}{nx}$

(A) $S_{10} = \cot^{-1}\frac{1+11x^2}{10x} = \tan^{-1}\frac{10x}{1+11x^2}$

$= \pi/2 - \tan^{-1}\frac{1+11x^2}{10x}$

(B) $\lim_{n\to\infty} \cot\left(\cot^{-1}\frac{1+(n+1)x^2}{nx}\right) = x$

(C) $S_n = \cot^{-1}\frac{1+4x^2}{3x} = \pi/4$

$\frac{1+4x^2}{3x} = 1 \Rightarrow 1 + 4x^2 - 3x = 0 \Rightarrow D < 0$

(D) $\tan(s_n(x) = \frac{nx}{1+(n+1)x^2}$

Since $\frac{1+(n+1)x^2}{2} \geq \sqrt{(n+1)}\,.\,x \geq nx$

$\frac{1+(n+1)x^2}{\sqrt{n+1}\,x} \geq 2$

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$\frac{\sqrt{n+1}\,x}{1+(n+1)x^2} \le 1/2$ as $n > 0$ & $n > 1$

$\frac{(n+1)x}{1+(n+1)x^2} \le \frac{\sqrt{n+1}}{2}$

Hence, D is wrong

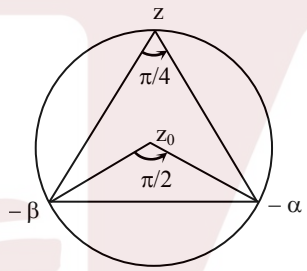
Q.16 For any complex number $w = c + id$, let $\arg(w) \in (-\pi, \pi]$, where $i = \sqrt{-1}$. Let α and β be real numbers such that for all complex numbers $z = x + iy$ satisfying $\arg\left(\frac{z+\alpha}{z+\beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle $x^2 + y^2 + 5x - 3y + 4 = 0$

Then which of the following statements is (are) **TRUE**?

(A) $\alpha = -1$ (B) $\alpha\beta = 4$ (C) $\alpha\beta = -4$ (D) $\beta = 4$

Ans. **[B,D]**

Sol.



$\text{Arg}\left(\frac{z_0+\alpha}{z_0+\beta}\right) = \pi/2$

$\frac{z_0+\alpha}{z_0+\beta} = i$

$-\frac{5}{2}+\frac{3}{2}i + \alpha = -\frac{5}{2}i - \frac{3}{2} + \beta i$

$z_0 = -\frac{5}{2} + \frac{3}{2}i$

$-1 + 4i = -\alpha + \beta i$

So $\alpha = 1$ & $\beta = 4$

SECTION – 4

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each quesiton, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numberic keypad in the place-desiginated to enter the answer.
- Answer to each question will be evaluated <u>according to the following marking scheme</u> :

Full Marks	: **+4**	If ONLY the correct integer is entered.
Zero Marks	: **0**	In all other cases.

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Q.17 For $x \in R$, the number of real roots of the equations $3x^2 - 4|x^2 - 1| + x - 1 = 0$ is .....

Ans. **[4]**

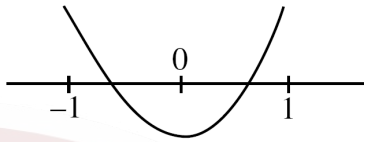
Sol. $3x^2 + x - 1 = 4|x^2 - 1|$

If $x \in [-1, 1]$,

$3x^2 + x - 1 = -4x^2 + 4 \Rightarrow 7x^2 + x - 5 = 0$

say $f(x) = 7x^2 + x - 5$

$f(1) = 3$; $f(-1) = 1$; $f(0) = -1$



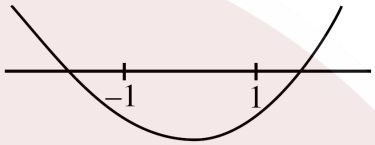
[Two Roots]

If $x \in (-\infty, -1] \cup [1, \infty)$

$3x^2 + x - 1 = 4x^2 - 4 \Rightarrow x^2 - x - 3 = 0$

Say $g(x) = x^2 - x - 3$

$g(-1) = -1$; $g(1) = -3$



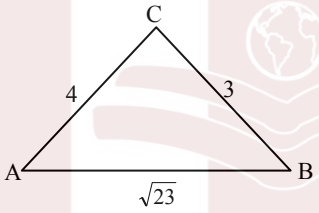
[Two Roots]

So total 4 roots.

Q.18 In a triangle ABC, let AB = $\sqrt{23}$, BC = 3 and CA = 4. Then the value of $\frac{\cot A + \cot C}{\cot B}$ is .....

Ans. **[2]**

Sol.



$$\frac{\cot A + \cot C}{\cot B} = \frac{\frac{\cot A}{a} + \frac{\cot C}{c}}{\frac{\cot B}{b}}$$

$$= \frac{bc\cot A + ab\cot C}{ac\cot B}$$

$$= \frac{b^2 + c^2 - a^2 + a^2 + b^2 - c^2}{a^2 + c^2 - b^2}$$

$$= \frac{2(b^2)}{a^2 + c^2 - b^2} = \frac{2.16}{a + 23 - 16} = \frac{2.16}{16}$$

$= 2$

Q.19 Let $\vec{u}$, $\vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and $\vec{u} \cdot \vec{w} = 1$, $\vec{v} \cdot \vec{w} = 1$, $\vec{w} \cdot \vec{w} = 4$.

If the volume of the parallelopiped , whose adjacent sides are represented by the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$, is $\sqrt{2}$, then the value of $|3\vec{u} + 5\vec{v}|$ is .....

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Ans. **[7]**

Sol. $\left[\vec{u}\ \vec{v}\ \vec{w}\right]^2 = \left(\sqrt{2}\right)^2$

$$\begin{vmatrix} \vec{u}\cdot\vec{u} & \vec{u}\cdot\vec{v} & \vec{u}\cdot\vec{w} \\ \vec{u}\cdot\vec{v} & \vec{v}\cdot\vec{v} & \vec{v}\cdot\vec{w} \\ \vec{u}\cdot\vec{w} & \vec{u}\cdot\vec{w} & \vec{w}\cdot\vec{w} \end{vmatrix} = 2$$

$$\begin{vmatrix} 1 & \vec{u}\cdot\vec{v} & 1 \\ \vec{u}\cdot\vec{v} & 1 & 1 \\ \vec{u}\cdot\vec{w} & 1 & 4 \end{vmatrix} = 2$$

$$(4-1) - \vec{u}\cdot\vec{v}\,(4\vec{u}\cdot\vec{v} - 1) + (\vec{u}\cdot\vec{v} - 1) = 2$$

$$2 + 2\vec{u}\cdot\vec{v} - 4\,(\vec{u}\cdot\vec{v})^2 = 2$$

$$2(2\vec{u}\cdot\vec{v} - 1)\,(\vec{u}\cdot\vec{v}) = 0$$

$$\vec{u}\cdot\vec{v} = \frac{1}{2} \text{ as } \vec{u}\cdot\vec{v} \neq 0s$$

Now $|3\vec{u} + 5\vec{v}|^2 = 9 + 25 + 30\,\vec{u}\cdot\vec{v}$

$= 9 + 25 + 15 = 49$

So $|3\vec{u} + 5\vec{v}| = 7$

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